MIL-F-25880B(USAF)
21 MARCH 1972
SUPERSEDING
MIL-F-25880A(USAF)
29 January 1960

MILITARY SPECIFICATION

FILTER, BAND PASS, BAND SUPPRESSION F-339/A

1. SCOPE

1.1 Scope. - This specification covers the general requirements for one type of band pass, band suppression filter, designated Filter, Band Pass, Band Suppression F-339/A.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

Military

MIL-C-3643	Connectors, Coaxial, Radio Frequency, Series HN
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-P-17555	Preparation for Delivery of Electronic Equipment; Miscellaneous Electrical Equipment (Except Rotating Electrical Equipment) and Associated Repair Parts
MIL-C-45662	Calibration System Requirements

STANDARDS

Military

MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-831	Test Reports, Preparation of

DRAWINGS

Air Force

59C12689	Installation Information - Filter, Bandpass-Band Suppression F-339/A and Plate, Mounting, Filter MT-1904/A
59D12831	Plate, Mounting, Filter MT-1904/A - Assembly of

(Copies of documents required by contractors in connection with specific procurement functions shall be obtained from the procuring activity or as directed by the contracting officer.)

FSC 5915

Source: https://assist.dla.mil -- Downloaded: 20260-20T07:26Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Preproduction.- This specification makes provisions for First Article testing.

3.2 General Specification.- The requirements of MIL-E-5400 for class 2 equipment apply as requirements of this specification with the exceptions and additions called out herein. When the two specifications conflict, this specification shall govern.

3.2.1 Design Objective.- Minimum size and weight, simplicity of operation, and improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives in the design of the equipment specified herein. The use of materials parts, and processes other than those required by MIL-E-5400 shall be investigated and when it appears that a substantial reduction in size and weight or improvement in simplicity of design, performance, or reliability can be realized by their use, a request for approval of a non-standard part shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

3.2.2 Components.- The filter unit shall consist of the following:

Filter, Band Pass, Band Suppression F-339/A
Plate, Mounting, Filter MT-1904/A

3.3 Design and Construction.- The filter shall be designed and constructed to provide facilities for simultaneous operation of TACAN and IFF equipments connected to a common antenna system without interference, one with the other.

3.3.1 Mounting Provision.- The filter shall be equipped with a mount, MT-1904/A, in accordance with Drawing 59D12831.

3.3.2 Sealing.- The filter shall be hermetically or fusion sealed in a metal container. The design of the filter shall be such that all adjustments can be made internally during assembly without the need for any external adjustment.

3.3.3 Connectors.- The filter shall be equipped with a pressure-sealed type "HN" radio-frequency connector, suitable for use with Radio Frequency Plug UG-59E/U in accordance with MIL-C-3643/1. The plug arrangement shall be in accordance with Drawing 59C12689.

3.4 Electrical Performance

3.4.1 Frequency.- The filter shall meet the band pass and band suppression requirements specified under insertion loss herein for the frequency range from 950 to 1215 MHz.

3.4.2 Voltage Standing Wave Ratio

3.4.2.1 Antenna.- The voltage standing ratio (VSWR), measured at the ANT connector of the filter with the IFF and TACAN filter connectors terminated in 50-ohm resistive loads, shall not exceed 1.6:1 when measured over the frequency range of 950 to 1215 MHz.

Source: https://assist.dla.mil -- Downloaded: 202610-20T07:26Z
Check the source to verify that this is the current version before use.

3.4.2.2 TACAN.- The VSWR, measured at the TACAN coaxial input connector of the filter with the output connectors of the filter terminated in 50-ohm resistive loads, shall not exceed 1.6:1 when measured over the frequency range of 950 to 1215 MHz.

3.4.2.3 IFF.- The VSWR, measured at the IFF coaxial input connector of the filter with the output coaxial connectors of the filter terminated in 50-ohm resistive loads, shall not exceed 1.4:1 when measured over the IFF frequency bands of 1030 plus or minus 3 MHz and 1090 plus or minus 3 MHz.

3.4.3 Insertion loss.- The filter insertion loss between antenna and TACAN and between antenna and IFF shall be as shown in figure 1.

3.4.4 Isolation IFF to TACAN.- The filter shall provide sufficient isolation between IFF and TACAN signals to permit simultaneous operation of the two systems on one antenna; in no case shall the isolation be less than 15 db.

3.4.5 Radio-Frequency Power.- The filter shall be capable of handling a peak power input of 2.5 kw, 3 watts average, to either TACAN or IFF connectors at any frequency between 950 and 1215 MHz.

3.5 Size.- The outside dimensions of the filter, including mount, shall not exceed those specified in Drawing 59C12689.

3.6 Weight.- The weight, including mounting provisions, shall not exceed 3 pounds.

3.7 Identification of Product.- Identification marking of the filter shall be in accordance with MIL-STD-130 and shall contain the following information:

Nomenclature
Stock number
Serial number
Manufacturer's name or code
Manufacturer's part number

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of Tests.- The inspection and testing of the filter shall be as follows:

a. First Article . . . . . See 4.3.

b. Quality conformance inspection. . . . . See 4.4.

4.2 Standard Test Conditions.- Unless otherwise specified, the equipment shall be subjected to the tests under the following conditions:

a. Temperature: Room ambient, 25°C plus 15°C or minus 5°C.

b. Altitude: Normal ground.

c. Vibration: None.

d. Humidity: Room ambient up to 90 percent relative humidity.

Source: https://assist.dla.mil -- Downloaded: 202640-20T07:26Z
Check the source to verify that this is the currentversion before use.

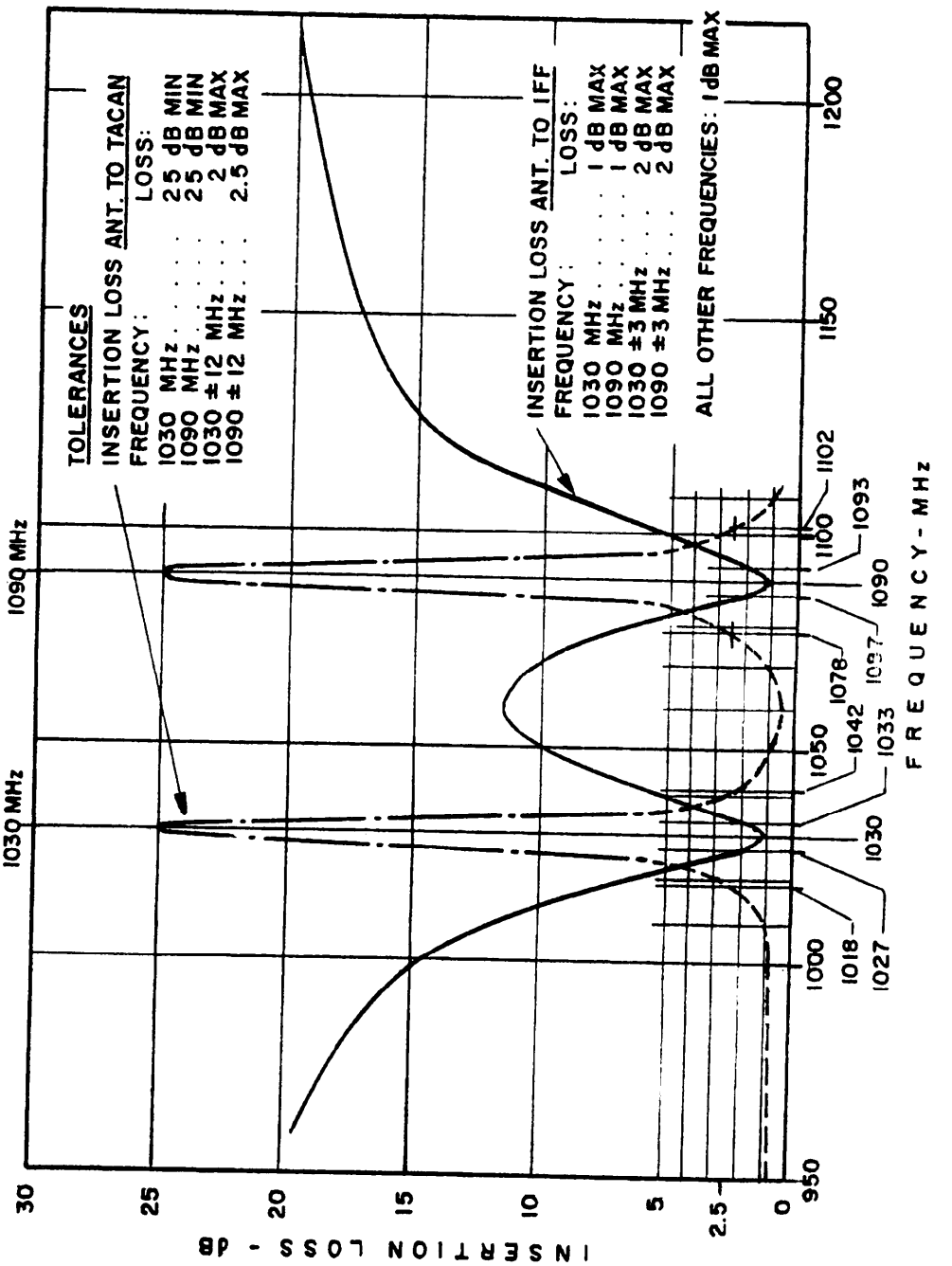


FIGURE 1 Insertion loss requirements

Source: https://assist.dla.mil -- Downloaded: 202640-20T07:26Z
Check the source to verify that this is the current version before use.

4.2.1 Test Equipment Calibration.- The test equipment used in performing the electrical tests specified herein shall comply with the applicable requirements of MIL-C-45662.

4.3 First Article

4.3.1 First Article Test Sample.- When specified in the contract, a minimum of 3 filters shall be subjected to the First Article tests specified herein.

4.3.2 First Article Test Report.- After completion of the First Article tests, the contractor shall prepare a test report in accordance with MIL-STD-831 and submit three copies to the procuring activity along with the First Article samples that have been tested.

4.3.3 Tests.- The First Article tests shall include the following tests and all of the tests described under Quality Conformance Inspection.

4.3.3.1 Radio-Frequency Power Handling Tests.- The filter shall be capable of handling, without damage, the power specified in 3.4.5 applied to the TACAN input terminal at frequencies of 970, 1030, 1090 and 1200 MHz, with the antenna terminated in a 2:1 VSWR load and the IFF terminal unterminated. The filter shall also be capable of handling, without damage, the power specified in 3.4.5 applied to the IFF terminal at 1090 MHz, with the antenna terminated in a 2:1 VSWR load and TACAN terminals unterminated. Each test shall be of 10 minute duration.

4.3.3.2 VSWR Test.- The VSWR produced by the filter shall be measured in accordance with figure 2 by use of conventional slotted line techniques. The measurements shall be performed for the ANT, IFF, and TACAN filter channels utilizing the following frequencies:

IFF	TACAN	ANT
1027 MHz	950 MHz	950 MHz
1030 MHz	1000 MHz	1000 MHz
1033 MHz	1018 MHz	1018 MHz
1087 MHz	1042 MHz	1027 MHz
1090 MHz	1060 MHz	1030 MHz
1093 MHz	1078 MHz	1042 MHz
	1102 MHz	1060 MHz
	1150 MHz	1078 MHz
	1215 MHz	1087 MHz
		1090 MHz
		1093 MHz
		1102 MHz
		1150 MHz
		1215 MHz

a. Test No. 1 (VSWR, ARN-21 channel), connect slotted line terminal A to ARN-21 filter connector. Terminate the IFF and ANT filter terminals with 50-ohm dummy loads. Proceed with normal slotted line VSWR measurements. (See figure 2).

b. Test No. 2 (VSWR, IFF channel), connect slotted line terminal A to IFF filter connector. Terminate the ARN-21 and ANT filter terminals with 50-ohm dummy loads. Proceed with normal slotted line VSWR measurements. (See figure 2.)

Source: https://assist.dla.mil -- Downloaded: 202640-20T07:26Z
Check the source to verify that this is the currentversion before use.

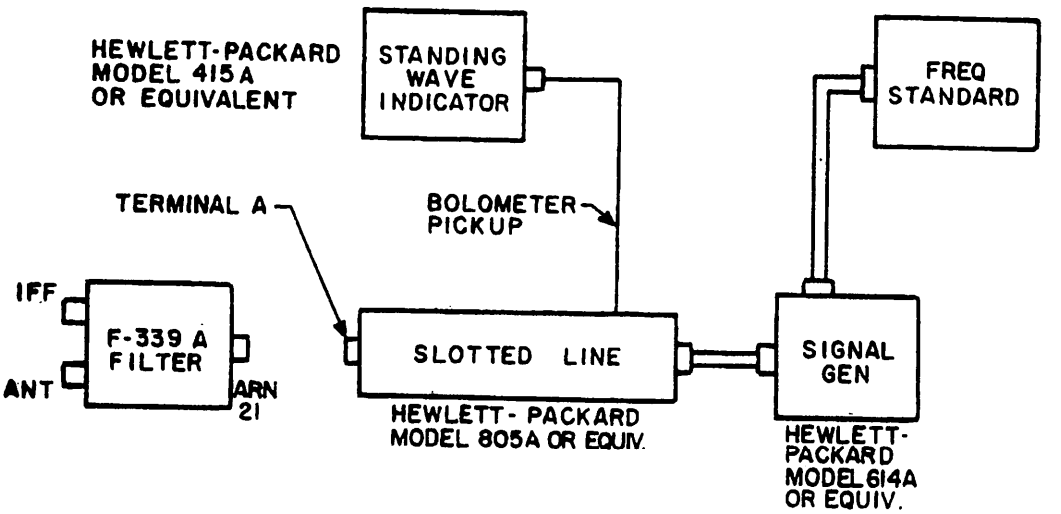


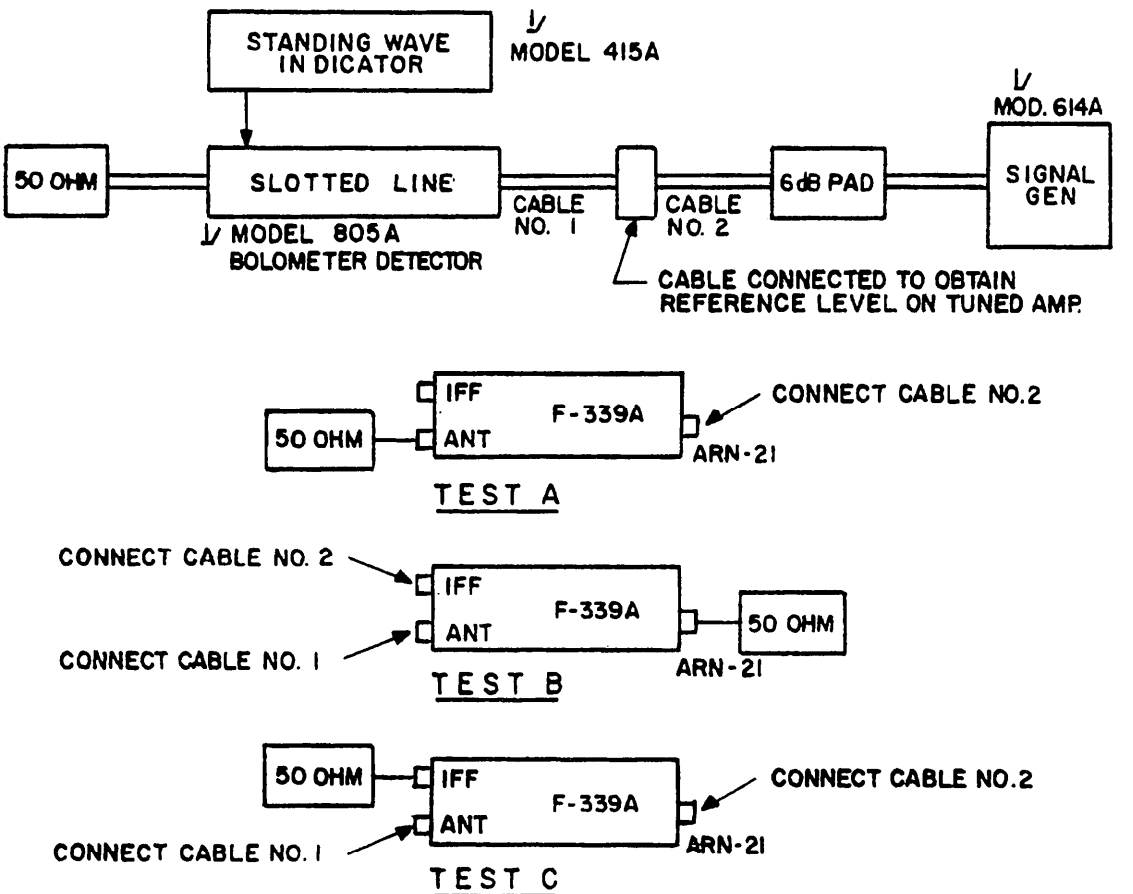
FIGURE 2. VSWR measurement setup

c. Test No. 3 (VSWR, ANT input), connect slotted line terminal A to ANT filter connector. Terminate the ARN-21 and IFF filter terminals with 50-ohm dummy loads. Proceed with normal slotted line VSWR measurements. (See figure 2.)

4.3.3.3 Insertion Loss.- The insertion loss introduced by the filter shall be measured for the antenna to IFF filter channel and the antenna to TACAN filter channel utilizing the procedure shown in figure 3. The following frequencies shall be used in the performance of both the TACAN and IFF measurements:

IFF	TACAN
950 MHz	950 MHz
1000 MHz	1000 MHz
1027 MHz	1018 MHz
1030 MHz	1030 MHz
1033 MHz	1042 MHz
1050 MHz	1050 MHz
1060 MHz	1078 MHz
1070 MHz	1090 MHz
1087 MHz	1102 MHz
1090 MHz	1150 MHz
1093 MHz	1215 MHz
1120 MHz	
1150 MHz	
1215 MHz	

Source: https://assist.dla.mil -- Downloaded: 202640-20T07:26Z
Check the source to verify that this is the currentversion before use.



1/ HEWLETT-PACKARD MODEL NUMBERS OR EQUIVALENT.

FIGURE 3. Isolation and insertion loss measurement setup

Source: https://assist.dla.mil -- Downloaded: 202640-20T07:26Z
Check the source to verify that this is the currentversion before use.

a. Connect cables No. 1 and No. 2 together as shown in figure 3 to obtain reference level on slotted line measurement equipment. Repeat this measurement prior to each test.

b. Test A (IFF to TACAN isolation measurement), connect cables No. 1 and No. 2 to Filter F-339/A as shown in figure 3, with 50-ohm dummy load connected to ANT terminal. Proceed with normal slotted line measurements to determine attenuation.

c. Test B (insertion loss ANT-IFF), connect cables No. 1 and No. 2 to filter as shown in figure 3, with 50-ohm dummy load connected to ARN-21 terminal. Proceed with normal slotted line measurements to determine attenuation.

d. Test C (insertion loss ANT/ARN-21), connect cables No. 1 and No. 2 to filter as shown in figure 3, with 50-ohm dummy load connected to IFF terminal. Proceed with normal slotted line measurements to determine attenuation.

4.3.3.4 Isolation Test. - The amount of isolation provided by the filter between TACAN and IFF, as measured at the TACAN and IFF coaxial input terminals, shall be measured in accordance with figure 3, utilizing the following frequencies:

FREQUENCY MHz

950	1078
1000	1087
1018	1090
1027	1092
1030	1102
1033	1150
1042	1215
1060	

Test methods other than those shown in figure 3 must be approved by the procuring activity.

4.3.3.5 Environmental Tests.- The filter shall be subjected to each of the environmental tests of MIL-E-5272 listed below. The filter shall be operated during all tests except the salt spray and shock tests. The filter shall meet all the requirements of this specification during and after the environmental tests:

Vibration	Procedure XII (1/2 time at 95°C)
High temperature test	Procedure II
Low temperature test	Procedure II
Temperature-shock test	Procedure I
Altitude test	Procedure VI (condition D)
Salt spray test	Procedure I
Shock test	Procedure V

4.4 Quality conformance Inspection.- Quality conformance inspection shall consist of:

a. Individual tests.
b. Sampling tests.

Source: https://assist.dla.mil -- Downloaded: 20260-20T07:26Z
Check the source to verify that this is the current version before use.

4.4.1 Individual Tests.- Individual test shall be conducted on every item of equipment submitted for acceptance under the contract. The individual tests shall consist of the following tests:

4.4.1.1 Examination of Product.- Each part or assembly produced under this specification shall be subjected to such inspection and testing as necessary to determine compliance with the materials, workmanship, finishes, and mechanical detail requirements of this specification.

4.4.1.2 VSWR Quality Conformance Test.- The VSWR test shall be measured as specified in 4.3.3.2 with the exception that the following frequencies shall be used:

TACAN	ANT	IFF
950 MHz	950 MHz	1027 MHz
1018 MHz	1018 MHz	1030 MHz
1042 MHz	1027 MHz	1033 MHz
1078 MHz	1030 MHz	1087 MHz
1102 MHz	1033 MHz	1090 MHz
1215 MHz	1042 MHz	1093 MHz
	1078 MHz	
	1087 MHz	
	1090 MHz	
	1093 MHz	
	1102 MHz	
	1215 MHz	

4.4.1.3 Insertion Loss.- The insertion loss introduced by the filter shall be measured as specified in 4.3.3.3 with the exception that the following frequencies shall be used:

TACAN	IFF
950 MHz	1027 MHz
1018 MHz	1030 MHz
1030 MHz	1033 MHz
1042 MHz	1087 MHz
1078 MHz	1090 MHz
1090 MHz	1093 MHz
1102 MHz	
1215 MHz	

4.4.1.4 Immersion Test.- The filter shall be subject to an immersion test in accordance with MIL-E-5272, procedure I.

4.4.1.5 Isolation Test.- The amount of isolation provided by the filters between TACAN and IFF, as measured at the TACAN and IFF coaxial input terminals, shall be measured in accordance with figure 3, utilizing the following frequencies:

Source: https://assist.dla.mil -- Downloaded: 202610-20T07:26Z
Check the source to verify that this is the current version before use.

FREQUENCY MHz
1027
1030
1033
1087
1090
1093

4.4.2 Sampling Tests.- Two filters shall be selected at random from each quantity of 100 or fraction thereof on order and subjected to the following tests:

Vibration test
Temperature shock test
Measurements, as specified in 4.3.3.2, 4.3.3.3 and 4.3.3.4.

These tests shall be conducted before and after the environmental tests. The tests of 4.3.3.1 shall also be conducted before and after the environmental tests specified above.

5. PREPARATION FOR DELIVERY

5.1 General.- Preparation for delivery shall be in accordance with MIL-P-17555.

6. NOTES

6.1 Intended Use.- The filter covered by this specification is intended for use in USAF Aircraft to couple TACAN and IFF equipment to one antenna.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Number of First Article samples, point of testing, and delivery of First Article test report (see 4.3.2).

c. Selection of applicable levels of preservation-packaging, and packing.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:
Air Force - 80

User activity:
Air Force - 11

Preparing activity:
Air Force - 80

(Project 5915-F024)

Source: https://assist.dla.mil -- Downloaded: 20260-20T07:26Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

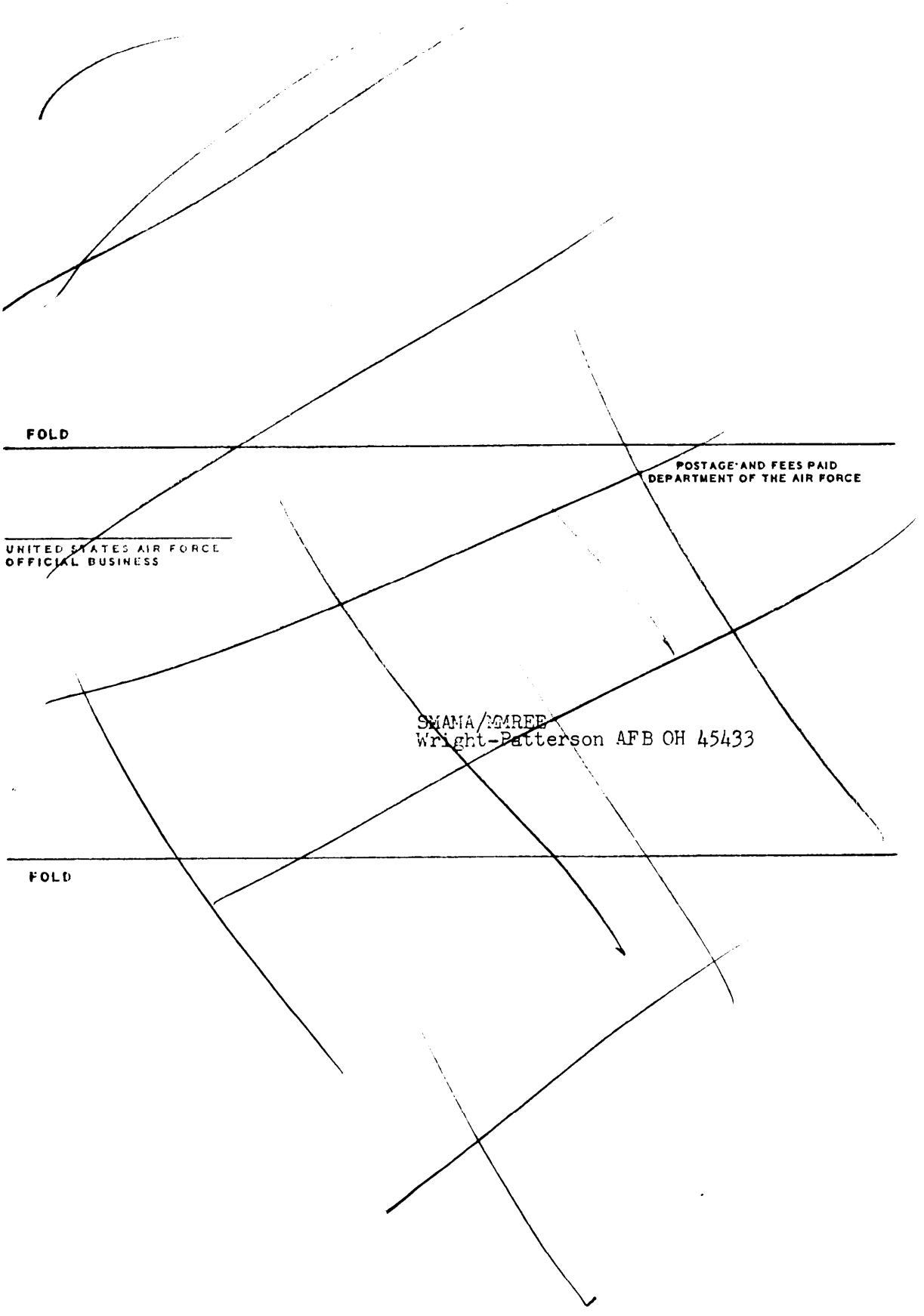
DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

AFLC–WPAFB–OCT 67 2M

Source: https://assist.dla.mil -- Downloaded: 2026l0-20T07:26Z
Check the source to verify that this is the currentversion before use.

determine compliance with the requirements these tests shall



Source: https://assist.dla.mil -- Downloaded: 20261-20T07:26Z
Check the source to verify that this is the current version before use.